

# Yogesh Chebrolu

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## SUMMARY

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AI Software Engineer with 1 year of experience building production-grade AI agents, RAG pipelines, evaluation systems, and multimodal applications. Strong backend experience with Python, FastAPI, TypeScript, Next.js, AWS, PostgreSQL, Redis, and event-driven workflows. Experienced in building LLM-powered automation systems using LangChain, LangGraph, OpenAI/Gemini APIs, vector databases, tool calling and parallel agents.

## EXPERIENCE

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### Kairos Computer - AI Engineering Intern

July 2025 - Present

- Built an LLM-powered evaluation pipeline using Gemini 2.0 Flash on AWS Lambda that analyzes completed task runs in 15-message windows, classifies failures such as auth misses, tool/API errors, and incomplete inputs, tracks per-run token costs to Supabase, and alerts developers in Slack - cutting failure detection from manual review to minutes.
- Eliminated the platform's top failure mode by adding tool introspection (`get_tool_context`) to the user-facing agent so it collects required credentials and inputs before triggering automations, and built an AgentMail email channel that lets users trigger agents by CC'ing or forwarding Gmail threads with full web-experience parity.
- Rebuilt the communication agent from scratch in TypeScript/Next.js during the company's migration away from Python/FastAPI, stress-tested the browser agent against 500+ manual evaluation tasks, and drove GTM outreach across email, LinkedIn, and Reddit to 100+ contacts per day, converting 3 paying subscribers.

### Stealth Startup - Educational AI Platform | AI Research Intern

March 2025 - June 2025

- Contributed to an AI-powered education platform focused on personalized learning experiences for undergraduate students.
- Designed multimodal RAG workflows combining LLMs, retrieval, memory, intelligent search, and tool-use patterns.
- Developed MCP plugin for Claude Desktop to connect with YouTube Videos to get multimodal retrieval context.

## PROJECTS

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### Alpha Brain - AI-First Personal Knowledge Workspace | <http://yogeshuu.vercel.app/alpha-brain>

- Built Alpha Brain, an AI-first personal knowledge workspace that works across Web and Telegram that converts raw thoughts into structured ideas, custom categories, action deadlines, research-backed articles, and persistent conversations.
- Implemented the platform with Next.js, Hono, Convex, Vercel AI SDK, SSE streaming, tool calling, encrypted Telegram bot webhooks, and responsive chat UI, enabling user-specific conversation history, AI-assisted category/idea/article creation, and cross-platform continuity.
- Reduced the knowledge-management workflow from 5-7 manual steps to 1 prompt, cutting idea capture and organization time from 3-5 minutes to under 30 seconds per idea and eliminating manual idea/note categorization across separate tools.

### Cover Wisely - AI Insurance Recommendation Engine | <http://yogeshuu.vercel.app/covey>

- Built an LLM-based recommendation engine that scrapes and normalizes IRDAI-mandated CIS documents from 9+ insurers, enabling comparison across 300+ health and life policies.
- Built a retrieval-augmented extraction pipeline with per-policy Pinecone namespaces, Qwen3 embeddings, and grounded LLM extraction to structure 300+ policies into a 30-field schema at ~\$0.08/policy.

### Multimodal YouTube RAG Chrome Extension | [yogeshuu.vercel.app/youtube-rag](https://yogeshuu.vercel.app/youtube-rag)

- Built a Chrome side-panel assistant that lets users chat with any YouTube video using transcript, visual frame, screenshot, watchtime, and notes context.
- Implemented a multimodal RAG pipeline with React, TypeScript, FastAPI, Supabase, pgvector, multimodal embeddings, FFmpeg frame extraction, transcript chunking, and Gemini tool calling.
- Benchmarked retrieval across 6 video categories, achieving up to 28% Text Hit@5, 16% Image Hit@5, and approximately 480 aggregate Hit@5 matches across experiments.

**English-to-Italian Translator Using Transformers in PyTorch | [yogeshuu.vercel.app/transformer-pytorch](https://yogeshuu.vercel.app/transformer-pytorch)**

- Implemented a Transformer model from scratch in PyTorch with custom multi-head attention, positional encoding, optimized feed-forward layers, and an end-to-end training pipeline.

**TOOLS & TECHNOLOGIES**

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**Languages:** Python, TypeScript, JavaScript, SQL, HTML/CSS

**Backend & Data:** FastAPI, Hono, REST APIs, Pydantic, PostgreSQL, MySQL, Redis, Supabase, Convex, MongoDB

**AI / ML:** LangChain, LangGraph, OpenAI API, Gemini API, Vercel AI SDK, CrewAI, RAG, AI Agents, Tool Calling, NLP, Deep Learning, Prompt Engineering, PyTorch

**Retrieval / Cloud / DevOps:** Pinecone, Qdrant, embeddings, semantic search, AWS Lambda, EC2, ECS, ECR, SQS, GitHub Actions, CI/CD, Git, Docker, Bun, Turborepo

**Frontend & Developer Tools:** React, Next.js, Streamlit, Cursor, Chrome Extensions

**EDUCATION**

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|---------------------------------------------------------------------------------|---------------------|
| <b>RGUKT - RK Valley   B.Tech, Computer Science and Engineering   CGPA: 8.7</b> | June 2022 - 2026    |
| <b>RGUKT - RK Valley   Pre-University Course, MPC   CGPA: 9.8</b>               | January 2020 - 2022 |
| <b>Sri Chaitanya E.M School   SSC   GPA: 10.0</b>                               | March 2019 - 2020   |

**AWARDS & PARTICIPATION**

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- Qualified GATE 2025 - Data Science and Artificial Intelligence (DA).
- Presented an AI-powered postal automation idea at Smart India Hackathon 2024.

**CERTIFICATIONS**

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- Introduction to Deep Learning, IIT Ropar (NPTEL) - Top 5%.
- ChatGPT Prompt Engineering for Developers - DeepLearning.AI.
- Building Systems with ChatGPT API - DeepLearning.AI.
- Data Science: Machine Learning - edX; Intro to Machine Learning - Hitloop Learning Platform.